

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Mock Paper Set 4

Time: 2 hours

Paper reference **8MA0/01**

Mathematics
Advanced Subsidiary
PAPER 1: Pure Mathematics

You must have:
 Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 16 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Using the substitution $u = x^2$ or otherwise, solve

$$4x^4 - x^2 - 18 = 0 \quad x \in \mathbb{R}$$

(4)



DO NOT WRITE IN THIS AREA

Question 1 continued

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(Total for Question 1 is 4 marks)



2. The curve C has equation

$$y = 5x^3 + \frac{19}{2}x^2 - 10x - 12$$

- (a) Find $\frac{dy}{dx}$ giving your answer in simplest form.

(2)

Given that the point $P(-2, 6)$ lies on C ,

- (b) find the equation of the tangent to C at P .

Give your answer in the form $y = mx + c$ where m and c are constants.

(3)



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Question 2 continued

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(Total for Question 2 is 5 marks)



3.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Show that $\frac{4 - \sqrt{6}}{5\sqrt{2} + 4\sqrt{3}}$ can be written in the form

$$a\sqrt{2} + b\sqrt{3}$$

where a and b are integers to be found.

(3)



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Question 3 continued

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(Total for Question 3 is 3 marks)



4.

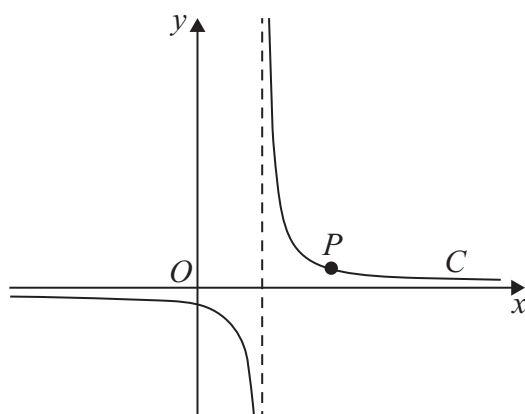
**Figure 1**

Figure 1 shows a sketch of part of the curve C with equation $y = f(x)$ where

$$f(x) = \frac{1}{x - 2}$$

(a) State

- (i) the equation of the asymptote of C that is parallel to the y -axis
- (ii) the coordinates of the point of intersection of C with the y -axis

(2)

The point $P(3, 1)$ lies on C , as shown in Figure 1.

(b) Find the point to which P is mapped when C is transformed to the curve with equation

- (i) $y = f(x) + 4$
- (ii) $y = f(2x)$

(2)



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Question 4 continued

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(Total for Question 4 is 4 marks)



- 5.** The curve C has equation $y = f(x)$ where

$$f(x) = 3x^2 - 12x + 17$$

- (a) Express $f(x)$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants.

(3)

- (b) Hence, or otherwise, state the coordinates of the minimum point of C .

(1)

- (c) Sketch the graph of C , showing clearly the coordinates of the point where the curve crosses the y -axis.

(2)



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6.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Solve the equation

$$27^{2x-3} = \frac{1}{9\sqrt{3}}$$

(3)



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Question 6 continued

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(Total for Question 6 is 3 marks)





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Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 7 marks)



8.

In this question you must show all stages of your working. Solutions relying on calculator technology are not acceptable.

$$f(x) = 5x^3 - 7x^2 - 8x + a$$

where a is a constant.

Given that $(x - 2)$ is a factor of $f(x)$

(a) show that $a = 4$

(2)

(b) Hence solve $f(x) = 0$

(4)



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Question 8 continued

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(Total for Question 8 is 6 marks)



9. Using the laws of logarithms, solve

$$\log_3(2x + 3) + \log_3(x - 4) = 3 + 2\log_3\sqrt{x}$$

(5)



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Question 9 continued

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(Total for Question 9 is 5 marks)



10.

In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.

Pam monitors the number of users of a social media website. The number of users, N million, is modelled by the equation

$$N = 17 + 25e^{0.08t}$$

where t is the time, in months, after monitoring began.

Using the model,

- (a) find the value of t when the number of users was 65 million, giving your answer to one decimal place,

(3)

- (b) find the rate of increase in the number of users with respect to time, exactly 2 **years** after monitoring began.
Give your answer to 3 significant figures.

(3)

- (c) State a limitation of the model.

(1)



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Question 10 continued

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(Total for Question 10 is 7 marks)



11. The binomial expansion, in ascending powers of x , of

$$\left(3 + \frac{k}{9}x\right)^7$$

where k is a **positive** constant, can be written in the form

$$A + Bx + Cx^2 + \dots$$

where A , B and C are constants.

(a) Find the value of A

(1)

Given that $B + C = 406$

(b) show that

$$9k^2 + 81k - 58 = 0$$

(3)

(c) Hence find

(i) the value of k

(ii) the value of C

(3)

(d) Find the first 2 non-zero terms, in ascending powers of x in the series expansion of

$$f(x) = \left(3 + \frac{k}{9}x\right)^7 + \left(3 - \frac{k}{9}x\right)^7$$

giving each term in simplest form.

(2)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

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(Total for Question 11 is 9 marks)



12.

In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$3 \sin \theta \tan \theta = 4$$

can be written as

$$3 \cos^2 \theta + 4 \cos \theta - 3 = 0 \quad (3)$$

(b) Hence solve, for $0 < x < 180^\circ$, $x \neq 45^\circ$, $x \neq 135^\circ$

$$3 \sin 2x \tan 2x = 4$$

giving your answers to one decimal place. (4)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Handwriting practice area with horizontal lines.

(Total for Question 12 is 7 marks)



13. Relative to a fixed origin O

- point A has position vector $\mathbf{i} - 3\mathbf{j}$
- point B has position vector $7\mathbf{i} + 5\mathbf{j}$

Given that the point C is such that $2\overrightarrow{AB} = \overrightarrow{BC}$

- (a) find the position vector of C .

(2)

Relative to O , point D has position vector $2\mathbf{i} + (p - 4)\mathbf{j}$ where p is a constant.

The points A , B and D form the triangle ABD .

Given that $|\overrightarrow{AD}| = \sqrt{17}$

- (b) find the largest possible size of angle DAB .
Give your answer in degrees to one decimal place.

(5)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Handwriting practice area with horizontal lines.

(Total for Question 13 is 7 marks)



14.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The circle C has equation

$$x^2 + y^2 - 4kx + 2ky + 80 = 0$$

where k is a constant, $k > 4$

(a) Find, in terms of k ,

(i) the coordinates of the centre of C

(ii) the radius of C

(3)

Given that the line with equation $y = k - 2x$ does **not** meet or touch C ,

(b) find the range of possible values of k , giving your answer in set notation.

(5)



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Question 14 continued

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Question 14 continued

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Question 14 continued

Handwriting practice area with horizontal lines.

(Total for Question 14 is 8 marks)



15.

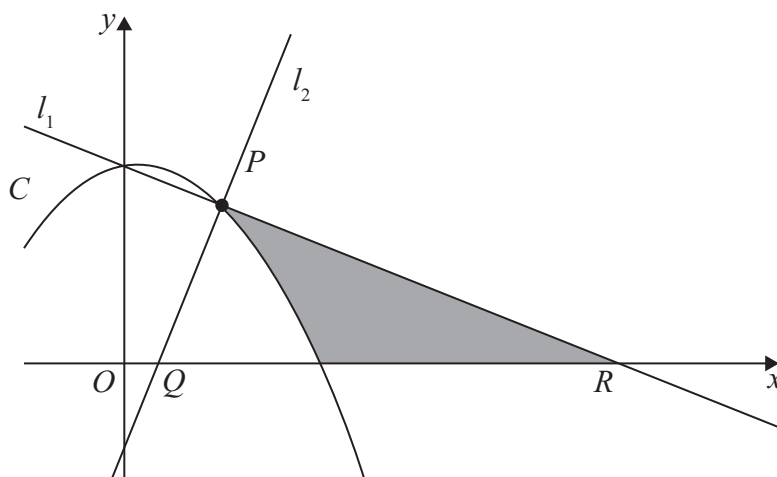


Figure 2

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 2 shows a sketch of part of the curve C with equation

$$y = -\frac{3}{50}x^2 + \frac{1}{5}x + 20$$

and a sketch of the line l_1 with equation

$$5y + 2x = 100$$

- (a) State the gradient of l_1 (1)

The point $P(10, 16)$ lies on C .

The line l_2 is perpendicular to l_1 and intersects l_1 at P as shown in Figure 2.

- (b) Find an equation for l_2 , giving your answer in the form $ax + by + c = 0$ where a , b and c are constants. (3)

Given that l_2 crosses the x -axis at the point Q

- (c) find the x coordinate of Q (2)

Given also that l_1 crosses the x -axis at the point R

- (d) find the exact area of triangle PQR (3)

The shaded region, shown in Figure 2, is bounded by C , the x -axis and l_1

- (e) Find, in simplest form, the fraction of triangle PQR that is shaded. (6)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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Question 15 continued

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(Total for Question 15 is 15 marks)



16. Prove, using algebra, that for all positive integers n ,

$$n^3 + 3n^2 - 2n$$

is always even.

(4)



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Question 16 continued

Lined area for writing the answer to Question 16.



Question 16 continued

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(Total for Question 16 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS

